

Dynamic Nelson-Siegel Yield Curve Modeling with Deep Learning Architecture

Yeonsoo Lim

Department of Applied Mathematics and Statistics
Johns Hopkins University

Advisor: Prof. Haoyang Cao

2025.12

1 Introduction

The estimation and forecasting of the yield curve, defined as the relationship between interest rates and the term to maturity of debt securities, is a central problem in financial economics. The yield curve serves as a compact representation of market expectations about future interest rates, inflation, and economic activity, and it plays a critical role for a wide range of economic agents. Fixed-income investors rely on yield curve forecasts for bond pricing, portfolio allocation, and interest rate risk management. Corporations use the yield curve as a benchmark for capital budgeting, debt issuance, and maturity choice. Central banks monitor the yield curve as an indicator of monetary policy transmission and macroeconomic expectations, while fiscal authorities use it to assess sovereign borrowing conditions and debt sustainability.

Despite its importance, the yield curve exhibits complex and time-varying dynamics. Under normal conditions, the curve is upward sloping, reflecting expectations of economic growth and inflation as well as term premia. However, its shape can change markedly in response to macroeconomic shocks and policy interventions. Yield curve inversions have historically preceded economic slowdowns and recessions, although their interpretation depends on the underlying policy regime. After the Global Financial Crisis of 2008, yield curve dynamics became more difficult to model due to unconventional monetary policies such as quantitative easing, zero interest rate policies, and large-scale asset purchases. These interventions altered the relationship between short- and long-term rates, weakened traditional arbitrage mechanisms, and challenged the assumptions underlying standard term structure models.

From a forecasting perspective, accurately predicting the entire yield curve, rather than individual maturities, is particularly important. Forecast errors at different maturities are highly correlated, and portfolio and policy decisions depend on the joint evolution of yields across the term structure. This motivates modeling approaches that capture cross-maturity dependence, nonlinear dynamics, and interactions with macro-financial variables. Traditional parametric models provide interpretability and theoretical grounding, but they may struggle to adapt to regime changes and nonlinear patterns observed in recent decades. These limitations have motivated growing interest in flexible, data-driven methods that can complement economic structure with modern machine learning techniques.

Against this background, this study investigates whether modern deep learning architectures can improve yield curve estimation and forecasting while preserving economic structure and in-

interpretability. Building on the Nelson–Siegel representation, the analysis evaluates a class of neural network models designed to learn latent yield curve factors and their dynamics directly from data. In particular, the paper focuses on Transformer-based architectures, which are well suited to capturing long-range temporal dependence and cross-maturity interactions, and compares their performance with more conventional deep learning benchmarks and traditional econometric models.

In addition, the study examines the role of macroeconomic and financial information in yield curve forecasting by incorporating observable variables such as inflation, real activity indicators, and stock market price index into the model inputs. Forecast accuracy is assessed across maturities and horizons to evaluate both level performance and robustness under different market conditions.

1.1 Related Literature

The academic literature on yield curve modeling can be broadly classified into equilibrium-based models, no-arbitrage models, and empirical factor-based approaches. Equilibrium models, such as Vasicek and Cox–Ingersoll–Ross type models, derive the term structure from assumptions about the dynamics of the short rate and investors’ preferences. No-arbitrage affine term structure models impose restrictions that ensure bond prices are free of arbitrage opportunities and are widely used for pricing interest rate derivatives. While these frameworks offer strong theoretical foundations, empirical evidence suggests that they often perform poorly in out-of-sample yield forecasting, particularly at medium and long horizons.

An influential empirical alternative is the Dynamic Nelson–Siegel (DNS) model introduced by Diebold and Li (2006)[1]. Building on the static Nelson–Siegel curve, the DNS framework represents the yield curve using three latent factors commonly interpreted as level, slope, and curvature. These factors evolve dynamically over time, typically following autoregressive processes, and the model can be cast in state-space form and estimated using the Kalman filter. The DNS model is attractive due to its parsimony, interpretability, and strong empirical forecasting performance relative to many theoretical term structure models. Subsequent work has extended the DNS framework by incorporating macroeconomic variables, showing that inflation, real activity, and monetary policy indicators contain predictive information for yield curve dynamics.

A key limitation of the DNS model is that it does not explicitly enforce no-arbitrage conditions. To address this issue, Christensen, Diebold, and Rudebusch proposed the Affine Arbitrage-Free Nelson–Siegel (AFNS) model[2], which embeds the Nelson–Siegel factor structure within an affine no-arbitrage framework. The AFNS model preserves the intuitive interpretation of the level, slope, and curvature factors while ensuring theoretical consistency with arbitrage-free pricing. Empirical studies suggest that AFNS models can improve cross-sectional fit and provide a coherent link between yield curve dynamics and term premia, although gains in forecasting accuracy relative to DNS are often modest.

More recently, advances in deep learning have motivated the application of neural networks to yield curve estimation and forecasting. Unlike linear state-space models, neural networks can capture nonlinear relationships, regime-dependent dynamics, and complex interactions across maturities. Several studies propose deep learning versions of the Nelson–Siegel framework, often referred to as Deep Learning Nelson–Siegel (DLNS)[3] models, in which neural networks are used to estimate latent factors, decay parameters, or their dynamics directly from observed yields. Empirical results indicate that DLNS models can outperform traditional DNS benchmarks, particularly at shorter horizons and during periods of heightened market stress.

Building on this line of research, recent work explores convolutional and recurrent neural network architectures to model the yield curve as a structured time series object. However, these architectures may still face limitations in capturing long-range temporal dependencies and cross-

maturity interactions. Transformer-based models, which rely on self-attention mechanisms, offer a promising alternative by allowing the model to learn flexible dependency structures across both time and maturities. Despite their success in other sequence modeling tasks, the application of Transformers to yield curve forecasting remains relatively underexplored.

This paper contributes to the literature by systematically evaluating Transformer-based architectures for yield curve forecasting within a Nelson–Siegel-inspired framework. In addition, the paper examines whether incorporating macroeconomic and market sentiment variables, such as inflation, industrial production, equity market indicators, and survey-based measures, improves predictive performance. By comparing Transformers with established deep learning benchmarks and traditional econometric models, this study aims to clarify the trade-offs between interpretability, flexibility, and forecasting accuracy in modern yield curve modeling.

2 Problem Setup

2.1 Dynamic Nelson-Siegel (DNS) Model

Traditionally, the Dynamic Nelson–Siegel (DNS) framework proposed by Diebold and Li (2006)[1], and extended in Diebold et al. (2006)[4], has been widely used for modeling and forecasting the term structure of interest rates. The DNS model provides a parsimonious yet flexible representation of the yield curve by decomposing it into three economically interpretable latent factors that evolve dynamically over time.

The key idea underlying the DNS framework is that yields observed at different maturities can be well approximated by a small number of common factors with maturity-specific loadings. Empirically, principal component analyses of yield curve data consistently show that most variation in yields can be explained by three dominant components, which naturally correspond to level, slope, and curvature. The Nelson–Siegel functional form imposes a structured shape on these factor loadings, allowing the factors to be interpreted economically rather than treated as purely statistical constructs. Specifically, the DNS model takes the following form:

$$\begin{aligned} y_t &= \Lambda(\lambda)f_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H), \\ f_t &= (I - A)\mu + Af_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q), \end{aligned}$$

In this setup:

- $y_t = (y_t(\tau_1), \dots, y_t(\tau_N))' \in R^{N \times 1}$ denotes the vector of observed spot rates (yields) at time t for N different maturities $\tau_1, \dots, \tau_N$.
- $f_t = (L_t, S_t, C_t)' \in R^{3 \times 1}$ is the vector of latent yield curve factors at time t , which correspond to the level (L_t), slope (S_t), and curvature (C_t) components.
- $\Lambda(\lambda) \in R^{N \times 3}$ is the Nelson-Siegel factor loading matrix, which maps the latent factors into the observable yield space based on the decay parameter λ .
- $\varepsilon_t \sim \mathcal{N}(0, H)$ is the vector of measurement errors with a diagonal covariance matrix $H \in R^{N \times N}$.
- f_t evolves according to a first-order vector autoregressive process (VAR(1)) with unconditional mean $\mu = (\mu_L, \mu_S, \mu_C)'$, autoregressive coefficient matrix $A \in R^{3 \times 3}$, and innovation noise $\eta_t \sim \mathcal{N}(0, Q)$, where $Q \in R^{3 \times 3}$ is a positive definite covariance matrix.

- $I \in R^{3 \times 3}$ denotes the identity matrix.

The factor loading matrix $\Lambda(\lambda)$ is explicitly given by:

$$\Lambda(\lambda) = \begin{bmatrix} 1 & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} - e^{-\lambda\tau_1} \\ 1 & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} - e^{-\lambda\tau_2} \\ \vdots & \vdots & \vdots \\ 1 & \frac{1-e^{-\lambda\tau_N}}{\lambda\tau_N} & \frac{1-e^{-\lambda\tau_N}}{\lambda\tau_N} - e^{-\lambda\tau_N} \end{bmatrix} \quad (1)$$

In the DNS model, the measurement equation expresses observed yields as linear combinations of the latent factors through the factor loading matrix $\Lambda(\lambda)$. The first column of $\Lambda(\lambda)$ consists of ones, implying that the level factor loads equally on all maturities. As a result, changes in the level factor shift the entire yield curve up or down in a parallel fashion. The second column captures the slope component, with loadings that are high at short maturities and decay monotonically toward zero as maturity increases. This structure allows the slope factor to represent the spread between short-term and long-term interest rates. The third column governs the curvature component, with loadings that are close to zero at very short and very long maturities but peak at intermediate horizons. Consequently, the curvature factor primarily affects the medium-term segment of the yield curve.

The decay parameter λ plays a central role in shaping the maturity profiles of the slope and curvature loadings. It determines how quickly the influence of these factors diminishes with maturity and, in particular, controls the location at which the curvature loading reaches its maximum. In empirical applications, λ is often fixed at a value that places the curvature peak at a medium-term maturity, such as two to three years, to improve estimation stability.

To capture the time-series dynamics of the yield curve, the DNS framework models the latent factors as a vector autoregressive process of order one. This transition equation allows the level, slope, and curvature factors to evolve gradually over time while exhibiting persistence and cross-factor interactions. The inclusion of an unconditional mean vector ensures that the factor process is mean-reverting, which is consistent with observed long-run behavior of interest rates.

Casting the DNS model in state-space form enables joint estimation of the latent factors and model parameters using the Kalman filter. The Kalman filter provides optimal linear estimates of the latent factors given observed yields under Gaussian assumptions and naturally accommodates measurement noise and missing observations. This framework also facilitates multi-step-ahead forecasting of the entire yield curve by iterating on the estimated factor dynamics and mapping predicted factors back into yield space through the Nelson–Siegel loadings.

Overall, the DNS model strikes a balance between economic interpretability and empirical performance. Its structured factor loadings impose economically meaningful restrictions, while its dynamic specification allows for flexible modeling of yield curve evolution over time. These properties make the DNS framework a natural benchmark for evaluating more flexible modeling approaches, including deep learning extensions introduced later.

2.2 Deep Learning Nelson-Siegel (DLNS) Model

While the DNS framework provides a parsimonious and interpretable representation of the yield curve, it relies on linear state-space dynamics and fixed functional forms to map latent factors into observed yields. These assumptions may limit the model’s ability to capture nonlinear relationships, regime-dependent behavior, and complex interactions between yields and macroeconomic variables

that have become increasingly relevant in recent decades. To address these limitations, this study adopts a deep learning extension of the Nelson-Siegel framework, referred to as the Deep Learning Nelson-Siegel (DLNS) model.

The core idea of the DLNS approach is to retain the economically interpretable Nelson-Siegel structure while replacing the linear factor extraction mechanism with a flexible nonlinear mapping learned from data. Rather than estimating latent factors through a linear Kalman filter, the DLNS model uses a neural network to infer the latent yield curve factors, namely level (L_t), slope (S_t), curvature (C_t), as well as the decay parameter (λ_t), directly from observed yield curves and auxiliary macroeconomic time series. In this setting, the neural network functions as a nonlinear feature extractor that summarizes complex patterns in the input data into a small number of economically meaningful latent variables.

This nonlinear mapping component is referred to as the *pre-Nelson-Siegel (pre-NS) layer*. Intuitively, the pre-NS layer learns how combinations of yields across maturities and time convey information about the underlying shape of the yield curve. Unlike linear models, which assume that the contribution of each maturity is constant and additive, deep learning architectures can capture nonlinear interactions, threshold effects, and time-varying relationships that may arise from monetary policy shifts or changing macro-financial conditions.

In this study, four neural network architectures are considered for implementing the pre-NS layer. Following Lee (2023)[3], CNN-RNN and CNN-LSTM architectures are used as benchmark deep learning models. In addition, two extended architectures are proposed: a CNN-Transformer hybrid and a Transformer-only encoder. These architectures differ in how they process cross-maturity and temporal information, allowing for a systematic comparison of modeling choices.

Convolutional neural networks (CNNs), although originally developed for image processing, can be adapted to one-dimensional convolutions for time series applications. In the context of yield curve modeling, one-dimensional CNN layers operate across the maturity dimension at a given time point. This enables the model to learn local patterns in the cross section of yields, such as relative movements between neighboring maturities, which are informative about slope and curvature dynamics. By stacking multiple convolutional layers, the network can extract increasingly abstract maturity-wise features while maintaining computational efficiency.

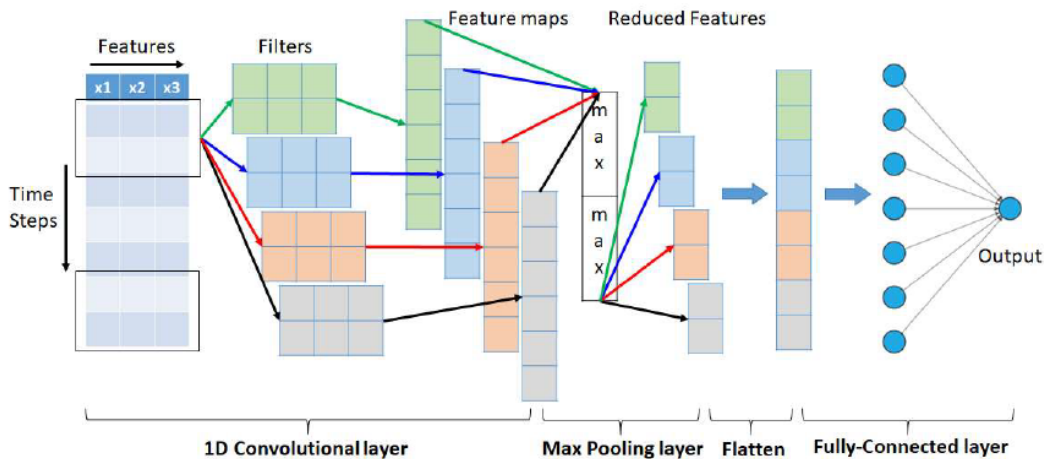


Figure 1: Structure of 1D CNN model[3]

To model temporal dependencies inherent in financial time series, the CNN-based feature extractors are combined with sequential architectures. Recurrent neural networks (RNNs) and long

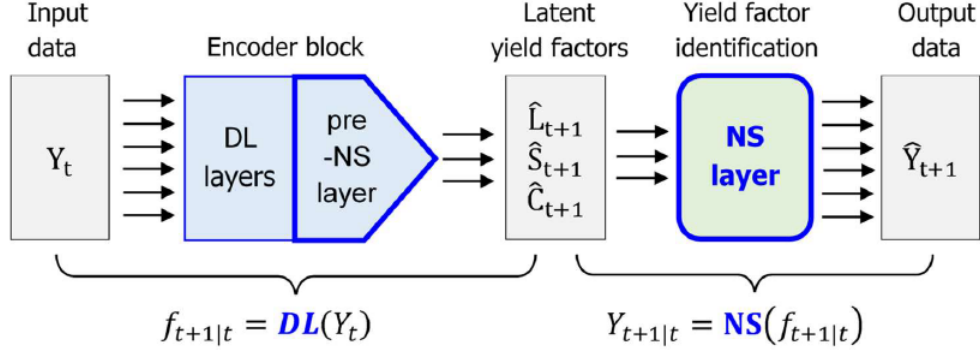


Figure 2: Structure of the Deep Learning Nelson-Siegel (DLNS) model proposed by Lee (2023)[3]

short-term memory (LSTM) networks capture persistence and dynamic evolution by propagating information forward through time. These models are well suited for learning gradual changes in latent yield factors, such as slow-moving level dynamics or cyclical slope behavior.

In contrast, the Transformer architecture relies on a self-attention mechanism that allows the model to directly relate observations across different time points without sequential recursion. This mechanism enables the Transformer to capture long-range temporal dependencies and global interactions more efficiently than traditional recurrent models. In the CNN-Transformer architecture, CNN layers first extract maturity-wise features, which are then processed by a Transformer encoder to model temporal relationships. In the Transformer-only architecture, self-attention is applied directly to the input sequences, allowing the model to jointly learn maturity-wise and temporal patterns within a unified framework.

Once the latent factors (L_t, S_t, C_t) and decay parameter λ_t are extracted by the pre-NS layer, they are passed to the *Nelson-Siegel layer*. This layer implements the Nelson-Siegel factor loading matrix $\Lambda(\lambda_t)$ using the standard parametric form. Importantly, this step preserves the economic interpretation of the model by ensuring that predicted yields are generated through level, slope, and curvature components with well-defined maturity loadings.

Finally, the output layer computes the predicted yield curve $\hat{y}_t$ as

$$\hat{y}_t = \Lambda(\lambda_t) \cdot f_t,$$

where $f_t = [L_t, S_t, C_t]^\top$. Both the Nelson-Siegel layer and the output layer are fully differentiable, allowing the entire model to be trained end-to-end via backpropagation. This design enables the model to jointly learn nonlinear feature extraction and economically structured yield curve reconstruction.

Figure 2 illustrates the overall DLNS architecture, while Figure 3 provides a detailed view of the Nelson-Siegel layer that maps latent factors into yields across maturities.

3 Dataset

3.1 Raw Data

The primary dataset for this study consists of monthly U.S. Treasury yield curve data sourced from the Federal Reserve's official repository: <https://www.federalreserve.gov/econres/feds/the-us-treasury-yield-curve-1961-to-the-present.htm>. This dataset is constructed based

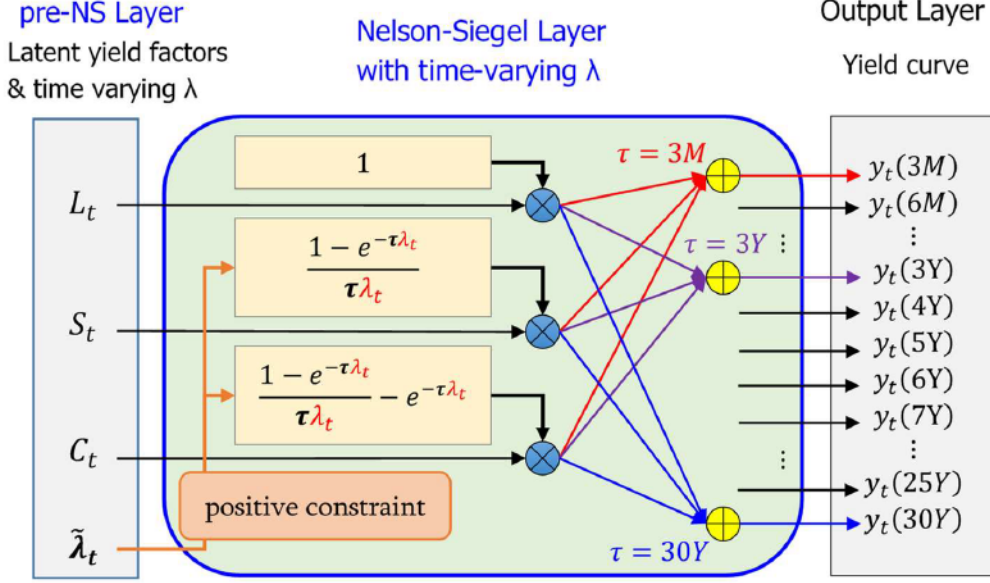


Figure 3: Internal structure of the Nelson-Siegel layer by Lee (2023)[3]

on the methodology described in the foundational paper by Gürkaynak, Sack, and Wright (2006) [5], which estimates smoothed zero-coupon yield curves for maturities ranging from 1 to 30 years. The data span the period from November 1985 to March 2025. To supplement this, yields for the 3-month and 6-month maturities were obtained from the Federal Reserve Economic Data (FRED) database and merged with the main dataset.

In addition to the yield curve data, the following macro-financial sentiment indicators are incorporated into the model:

- **NASDAQ Index:** Serves as a proxy for equity market performance and investor sentiment.
- **Industrial Production:** Captures real economic activity in the U.S. from the production side and reflects cyclical fluctuations in output.
- **Consumer Sentiment:** A monthly survey-based index published by the University of Michigan, providing forward-looking insights into household consumption trends, which constitute a major component of U.S. GDP.
- **Inflation:** Measured as the year-over-year percentage change in the U.S. Consumer Price Index (CPI). To account for potential lagged effects of inflation on the yield curve, the model incorporates CPI data with a one-month lag.

These variables aim to enhance the model's capacity to capture macroeconomic and financial sentiment influences on yield curve dynamics.

The dataset comprises a total of 472 monthly observations, spanning the period from 1985.12-2025.03. For the purposes of model development and evaluation, the data were partitioned into three distinct subsets. The training set encompasses 378 monthly observations (80.0% of the total), covering the period from 1985.12 to 2017.05. The validation set consists of 33 monthly observations (7.0% of the total), ranging from 2017.06 to 2020.02. Finally, the test set includes 39 monthly observations (13.0% of the total), covering the period from 2020.03-2025.03.

3.2 Data Preprocessing

The preprocessing steps applied are as follows:

1. **NASDAQ Index:** The raw NASDAQ index is first log-transformed to stabilize variance, followed by first differencing to obtain return series. The resulting return series is then standardized using z-score normalization.
2. **Industrial Production:** Seasonally unadjusted industrial production data is log-transformed to mitigate variance heteroskedasticity and account for exponential growth trends. The transformed series is subsequently standardized via z-score normalization.
3. **Consumer Sentiment:** The University of Michigan consumer sentiment index is also log-transformed to stabilize variance and then standardized using z-score normalization.
4. **Inflation:** The year-over-year change in the Consumer Price Index (CPI) is standardized using z-score normalization, consistent with the treatment of the other macro-financial indicators.
5. **Windowed Training Preparation:** The dataset is segmented into overlapping windows of W months to construct training samples. Each input sample has the shape $(W, M + K)$, where M denotes the number of maturities and K denotes the number of macro-financial indicators. The prediction target is the yield curve at the subsequent time step.

4 Methodology

This section describes the deep learning models employed, training design, and evaluation strategy for U.S. Treasury yield curve forecasting. I adopt the Deep Learning Nelson-Siegel (DLNS) framework proposed by Lee (2023) [3], extending it to include Transformer-based architectures. All models are implemented in PyTorch and trained using a windowed forecasting approach.

4.1 Model Architectures

This study implements four deep learning encoder variants as the pre-Nelson-Siegel (pre-NS) layer to extract the latent yield curve factors, namely level, slope, curvature, as well as the time-varying decay parameter λ . The four model variants are (i) CNN-RNN, (ii) CNN-LSTM, (iii) CNN-Transformer, and (iv) Transformer Only. While all models share a common Nelson-Siegel output structure, they differ in how cross-maturity and temporal information is processed in the encoder stage. Table 1 summarizes the architectural specifications and key hyperparameters for each variant.

Across the CNN-based models, the encoder architecture is designed in two stages. First, a convolutional block extracts cross-sectional features from the yield curve at each time point. Second, a sequential model captures temporal dependencies in these extracted features. This separation reflects the economic structure of the problem, in which yields at different maturities are jointly determined at each time point, while the evolution of the yield curve over time exhibits persistence and dynamic dependence.

The convolutional layers consist of one-dimensional filters applied along the maturity dimension. By using kernel sizes of length three and increasing the number of channels from 32 to 64, the CNN layers learn localized patterns across neighboring maturities, such as relative steepening or curvature changes in specific segments of the yield curve. The ELU activation function is used to introduce

Table 1: Model Architecture Specifications

structure	CNN-Transformer	CNN-RNN	CNN-LSTM	Transformer-Only
Input Processing				
Input Layer	(lookback window, input_dim)			
CNN Layers	Conv1D: out_channels=32→64, kernel_size=3			
CNN Activation	ELU			Self Attention
Pooling	MaxPool 1d (2×), stride=2			
Sequential Processing				
Sequential Model	Transformer	RNN (GRU)	LSTM	Transformer
Hidden Dimension	hidden_dim=64			
Attention Heads	nhead=4	N/A	N/A	nhead=4
Number of Layers	num_layers=2			
Feature Extraction				
Macro Integration	Concat			Projection
Final Projection	hidden_dim=64			
Final Activation	Tanh			
Output				
Pre-NS Layer	4 factors: [Level, Slope, Curvature, Lambda]			
NS Layer	Yield curve reconstruction with lambda constraint			

Notes: This table presents the architectural specifications of the four implemented deep learning models. The models were trained using the Adam (Adaptive Moment Estimation) optimizer with MSE loss function, a learning rate of 0.001, 100 training epochs, early stopping with a patience of 30, and a batch size of 16.

nonlinearity while maintaining stable gradient behavior. Max pooling with stride two is applied to reduce dimensionality and to focus the representation on salient cross-sectional features.

To avoid contaminating maturity-wise feature extraction with information that does not have a natural cross-sectional interpretation, the CNN block is applied exclusively to the yield curve. Macroeconomic variables are excluded at this stage and are instead incorporated after convolution. This design choice ensures that the CNN layers focus on extracting structural features of the yield curve itself, rather than mixing yield information with macro-financial signals prematurely.

After convolution, the extracted yield features are combined with macroeconomic variables and passed to a temporal encoder. In the CNN-RNN and CNN-LSTM models, this temporal processing is performed by recurrent neural networks and long short-term memory networks, respectively. These architectures summarize historical information through recursive state updates and are well suited for modeling persistent dynamics in latent yield factors, such as gradual movements in the level factor or cyclical changes in the slope.

In the CNN-Transformer model, temporal dependencies are captured using a Transformer encoder. The self-attention mechanism allows the model to weight past observations dynamically when forming representations at each time step. This enables the model to capture long-range dependencies and regime-dependent behavior more flexibly than recurrent architectures, which propagate information sequentially.

The Transformer Only model adopts a different design philosophy. Instead of separating cross-sectional and temporal processing, it applies a Transformer encoder directly to the full input sequence, including both yields and macroeconomic variables. Through self-attention, the model

jointly learns interactions across maturities and over time within a unified representation. Positional encodings are added to preserve the temporal ordering of observations, and causal masking is applied to ensure that predictions at time t depend only on information available up to that time.

For all four architectures, the output of the temporal encoder is projected into a fixed-dimensional feature space and passed through a nonlinear activation function. The final pre-NS layer produces four latent quantities, corresponding to the level, slope, curvature factors, and the decay parameter λ . These latent variables are then mapped into predicted yields through the Nelson-Siegel layer, which enforces the economically interpretable factor loading structure.

By maintaining a common Nelson-Siegel output layer while varying the encoder architecture, this setup allows for a controlled comparison of how different deep learning models extract latent yield curve factors. Differences in forecasting performance can therefore be attributed to the encoder’s ability to capture nonlinear cross-sectional and temporal patterns, rather than to differences in the yield curve reconstruction mechanism.

4.2 Pre-NS and Nelson-Siegel Layers

The encoder component, referred to as the pre-Nelson-Siegel (pre-NS) layer, serves as the interface between high-dimensional input data and the economically interpretable Nelson-Siegel representation. Given a sequence of observed yield curves and, when included, macroeconomic variables, the pre-NS layer produces a low-dimensional latent state consisting of four elements: the three Nelson-Siegel factors, namely level (L_t), slope (S_t), and curvature (C_t), as well as the decay parameter λ_t . This dimensionality reduction step is central to the DLNS framework, as it summarizes complex cross-sectional and temporal patterns in the data into a compact set of variables that directly characterize the shape of the yield curve.

Unlike the traditional DNS model, where latent factors are inferred through a linear state-space system estimated by the Kalman filter, the pre-NS layer relies on a nonlinear mapping learned from data. This mapping allows the model to capture nonlinear relationships between yields at different maturities, interactions with macroeconomic variables, and time-varying dynamics that are difficult to accommodate within linear Gaussian frameworks. At the same time, the output of the pre-NS layer is explicitly constrained to retain the economic interpretation of the Nelson-Siegel factors, ensuring that flexibility in factor extraction does not come at the cost of interpretability.

The latent variables produced by the pre-NS layer are then passed to the Nelson-Siegel (NS) layer, which transforms them into predicted yields across maturities. This transformation is governed by the Nelson-Siegel factor loading matrix $\Lambda(\lambda_t)$, which maps the level, slope, and curvature factors into the observable yield space in a maturity-specific manner. The predicted yield curve $\hat{y}_t$ is given by

$$\hat{y}_t = \Lambda(\lambda_t) \cdot f_t,$$

where $f_t = [L_t, S_t, C_t]^\top$. By explicitly enforcing this structure, the NS layer ensures that predicted yields are generated through economically meaningful channels rather than arbitrary nonlinear combinations.

A key modeling consideration in this layer concerns the decay parameter λ_t , which governs the shape of the factor loadings across maturities. In particular, λ_t determines the rate at which the slope factor decays and the maturity at which the curvature factor reaches its maximum influence. If unconstrained, the estimation of λ_t can lead to unstable or economically implausible factor loadings. To address this issue, and following the implementation in Lee (2023)[3], λ_t is constrained to lie within a predetermined interval that reflects prior economic knowledge about plausible curvature locations.

Specifically, the raw network output $\tilde{\lambda}_t$ is transformed using a scaled sigmoid function,

$$\lambda_t = \sigma(\tilde{\lambda}_t) \cdot \left(\frac{1.8}{24} - \frac{1.8}{60} \right) + \frac{1.8}{60},$$

which ensures that λ_t remains within the interval $[1.8/60, 1.8/24]$. This range corresponds to curvature peaks located between 60 and 24 months, a region that is commonly associated with medium-term movements in the yield curve. By imposing this constraint, the model maintains numerical stability and prevents the decay parameter from drifting toward extreme values that would distort the factor loading structure.

Importantly, both the NS layer and the final yield reconstruction are implemented using fully differentiable operations. This design allows the entire DLNS architecture, including the constrained decay parameter, to be trained end-to-end via gradient-based optimization. As a result, the model jointly learns nonlinear feature extraction in the pre-NS layer and economically structured yield curve reconstruction in the NS layer, without violating the theoretical constraints embedded in the Nelson-Siegel framework.

4.3 Training Objective and Data Handling

The DLNS models are trained to minimize the discrepancy between the predicted yield curve and the realized yield curve at the next time step. Specifically, the training objective is to minimize the mean squared error (MSE) between the one-step-ahead forecast $\hat{y}_{t+1}$ and the observed yield curve y_{t+1} . The loss function is defined as

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|y_{t+1}(\tau_i) - \hat{y}_{t+1}(\tau_i)\|^2,$$

where N denotes the number of maturities and τ_i indexes the maturity dimension. By aggregating errors across all maturities, this objective explicitly penalizes distortions in the overall shape of the yield curve rather than focusing on isolated points along the term structure.

A rolling window training scheme is employed to construct input-output pairs for supervised learning. At each time t , the model receives as input a fixed-length sequence of historical observations covering the previous W months and produces a forecast for the yield curve at time $t + 1$. This setup reflects a realistic forecasting environment in which only past and contemporaneous information is available when forming predictions.

Formally, each input sample is represented as a matrix of dimension $(W, M + K)$, where M denotes the number of yield maturities and K denotes the number of macroeconomic variables. The yield component of the input consists of the full cross section of yields at each time point, while the macroeconomic component includes observable indicators such as inflation and real activity measures. When CNN-based architectures are used, the yield and macroeconomic components are processed separately at early stages of the network, as described in the model architecture section.

To ensure numerical stability and efficient training, all input variables are standardized using statistics computed from the training sample only. Yield series are normalized maturity-wise, while macroeconomic variables are standardized individually to have zero mean and unit variance. This preprocessing step prevents variables with larger scales from dominating the loss function and facilitates convergence during optimization.

The dataset is split into training, validation, and test samples based on time ordering rather than random sampling, in order to avoid look-ahead bias. The training sample is used to estimate model parameters, the validation sample is used for hyperparameter tuning and early stopping, and

the test sample is reserved exclusively for out-of-sample performance evaluation. Early stopping is applied based on validation loss to mitigate overfitting, with model weights selected at the epoch that minimizes validation error.

All models are trained using gradient-based optimization with backpropagation, leveraging the fully differentiable structure of the pre-NS and Nelson-Siegel layers. This training strategy allows the model to jointly learn nonlinear feature extraction, latent factor dynamics, and yield curve reconstruction in an integrated framework that is consistent with the forecasting objective.

4.4 Evaluation Design and Recursive Forecasting

Out-of-sample evaluation is conducted using a holdout test period. For each model, recursive forecasting is performed over 1 to 6 month horizons. Specifically, given an input window of W months, the model predicts the next month’s yield curve, which is then appended to the input sequence to form the window for predicting the following month, and so on.

To investigate the model’s ability to capture long-term dependencies, I test two settings for the lookback window: $W = 6$ and $W = 9$ months. Transformer-based models are expected to outperform RNNs and LSTMs in the longer term.

5 Results

This section presents the empirical results of the deep learning Nelson-Siegel (DLNS) models evaluated in this study. The analysis focuses on four encoder architectures: CNN-Transformer, Transformer-Only, CNN-RNN, and CNN-LSTM. Model performance is assessed under multiple experimental configurations, with particular attention to forecasting accuracy across maturities and horizons, as well as the economic interpretability of the extracted latent factors. Unless otherwise stated, the benchmark specification considered here uses a lookback window of six months and excludes macroeconomic variables, allowing for a clean comparison of how different architectures extract information from yield curve data alone.

5.1 Forecasting Performance Overview

Forecasting performance is evaluated using root mean squared error (RMSE) computed over the entire yield curve. This metric captures both point-wise errors at individual maturities and distortions in the overall shape of the term structure. Table 2 reports RMSE values across forecast horizons ranging from one to six months, while Figure 4 visualizes the distribution of errors across maturities for the benchmark configuration.

Across all forecast horizons, the Transformer-Only model with a six-month lookback window and no macroeconomic inputs achieves the lowest RMSE. The performance advantage is particularly pronounced at short and medium horizons, where accurate modeling of cross-maturity interactions and persistent dynamics is most critical. The Transformer-Only architecture consistently outperforms both CNN-RNN and CNN-LSTM models and exhibits a modest but robust improvement relative to the CNN-Transformer hybrid.

These results suggest that explicitly separating cross-sectional and temporal feature extraction, as in CNN-based hybrids, is not strictly necessary for yield curve forecasting when self-attention mechanisms are employed. By jointly modeling dependencies across maturities and time, the Transformer-Only architecture appears better suited to capturing the global structure of yield curve dynamics, even with relatively short input windows.

The observed ranking of models is broadly consistent with the findings of Lee (2023)[3], who reports strong forecasting performance for DLNS models based on convolutional and recurrent networks. However, direct quantitative comparison is not feasible due to differences in the sample period, data preprocessing choices, and hyperparameter configurations. In addition, the absence of publicly available code in the original study limits reproducibility. Within the controlled experimental setup of this paper, the results indicate that Transformer-based architectures offer a meaningful improvement over recurrent alternatives for yield curve forecasting.

Figure 4 further illustrates that the performance gains of the Transformer-Only model are not confined to a narrow subset of maturities. Instead, improvements are observed across the term structure, with particularly strong gains at intermediate and long maturities, where forecasting errors tend to accumulate over longer horizons. This pattern suggests that the self-attention mechanism effectively captures long-range dependencies that are relevant for shaping the medium- and long-term segments of the yield curve.

Table 2: Forecasting results of DLNS models

Without Macro					With Macro			
lookback window = 6								
Forecast Horizon	CNN- Transformer	Transformer- only	CNN- RNN	CNN- LSTM	CNN- Transformer	Transformer- only	CNN- RNN	CNN- LSTM
1	36.4	35.5	39.8	50.8	43.0	41.4	45.7	53.8
2	48.5	46.4	52.4	63.2	56.0	55.2	62.0	63.3
3	57.8	54.5	63.1	73.3	65.4	65.8	77.1	72.0
4	64.0	59.9	71.8	81.9	72.3	73.8	90.5	80.6
5	69.5	64.1	80.7	89.1	77.2	79.9	103.1	89.5
6	74.7	67.7	89.6	95.4	78.8	83.4	113.5	95.9
lookback window = 9								
Forecast Horizon	CNN- Transformer	Transformer- only	CNN- RNN	CNN- LSTM	CNN- Transformer	Transformer- only	CNN- RNN	CNN- LSTM
1	42.3	40.5	38.6	46.4	43.4	46.6	58.9	67.7
2	55.6	51.9	49.7	62.5	54.8	58.5	79.4	82.1
3	66.8	61.7	58.6	75.4	64.2	65.9	98.7	94.4
4	75.2	69.4	65.7	86.9	71.2	71.9	115.9	105.4
5	82.7	76.4	72.0	98.0	77.2	78.5	132.2	115.8
6	89.4	83.0	78.0	108.5	79.5	83.3	145.3	125.6

Notes: RMSE is used as the evaluation metric. Bold values indicate the model with the lowest RMSE for each forecast horizon.

5.2 Dynamics of Latent Factors and Decay Parameter

Beyond forecasting accuracy, an important advantage of the DLNS framework lies in its ability to extract economically interpretable latent factors. Figure 5 plots the estimated Nelson-Siegel level, slope, curvature factors, and the time-varying decay parameter λ produced by the four model architectures under the benchmark configuration.

Overall, the extracted level and curvature factors exhibit broadly similar dynamics across models, indicating that these components are relatively robust to architectural choices. In contrast, the

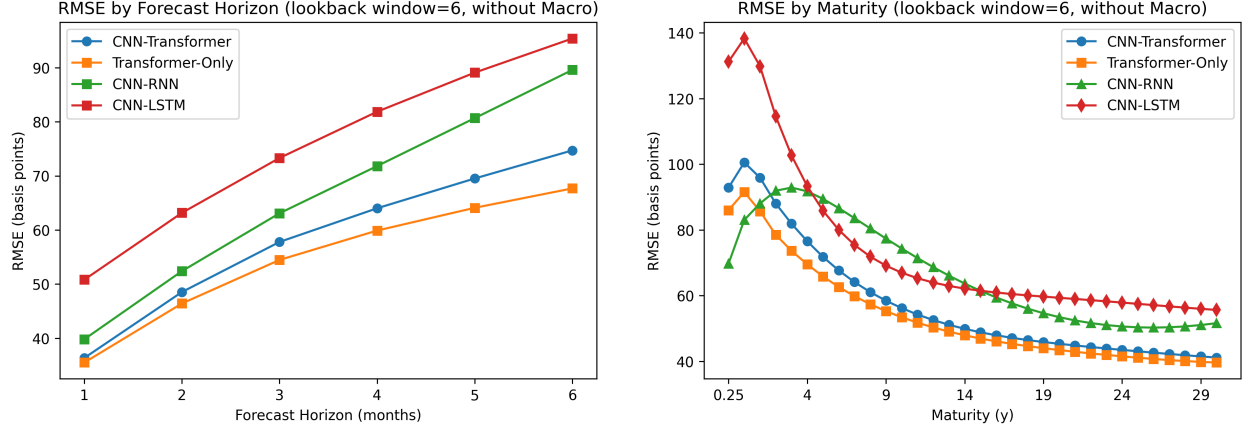


Figure 4: Forecasting Performance Comparison by Horizon and Maturity (Lookback Window = 6, Without Macro Inputs)

slope factor displays greater model-specific variation, particularly during periods associated with structural changes in monetary policy and financial markets. This sensitivity reflects the fact that the slope factor is closely tied to short-term rate dynamics and expectations about near-term policy actions, which are inherently more volatile and regime-dependent.

The level factor shows a gradual downward trend from the early 1990s through the pre-2008 period, consistent with the secular decline in nominal interest rates observed in U.S. Treasury markets. Following the Global Financial Crisis, the level factor drops sharply, coinciding with the implementation of zero interest rate policies and large-scale asset purchases by the Federal Reserve. The factor remains persistently low through the COVID-19 pandemic, reflecting prolonged accommodative monetary conditions, before rising sharply after 2022 in response to aggressive policy tightening aimed at containing elevated inflation.

The slope factor frequently enters negative territory after 2008, indicating recurrent yield curve inversions. These episodes are consistent with empirical observations during periods of unconventional monetary policy, subdued growth expectations, and heightened demand for long-term safe assets. Differences in the slope dynamics across models are most pronounced around turning points, suggesting that architectural choices influence how rapidly and smoothly the models adjust to shifts in the short end of the yield curve.

Taken together, these results highlight a key trade-off in DLNS modeling. While more flexible architectures such as Transformers deliver superior forecasting performance, they may also produce latent factor dynamics that differ subtly from those obtained using recurrent models. Nonetheless, the overall coherence of the extracted factors across architectures supports the interpretability of the DLNS framework and reinforces its usefulness as a tool for analyzing yield curve dynamics alongside forecasting performance.

5.3 Yield Curve Shape Prediction: Case Studies

To complement the aggregate forecasting metrics, this subsection examines the ability of the DLNS framework to reproduce the shape of the yield curve in specific economic environments. Figure 6 presents two case studies from the Transformer-Only model, comparing predicted and realized yield curves at two points in the test sample: October 2020 and March 2023. These episodes are chosen to contrast a relatively stable low-rate regime with a period of rapid monetary tightening and pronounced yield curve inversion.

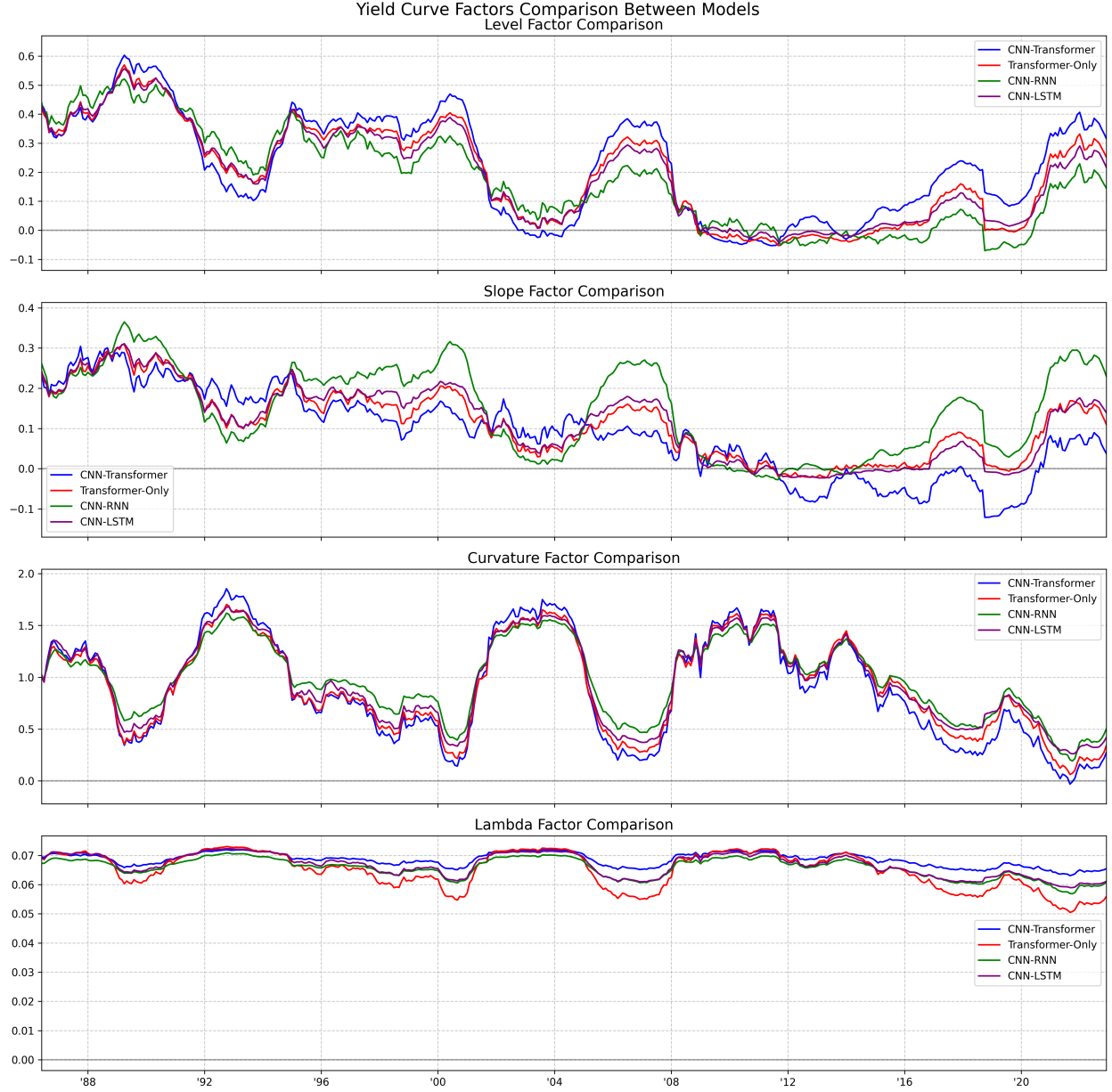


Figure 5: Comparison of Extracted Nelson-Siegel Factors Across Models (Lookback Window = 6, Without Macro Variables)

Panel (a) shows the forecast for October 2020, a period characterized by accommodative monetary policy and an upward-sloping yield curve. In this case, the model successfully captures the overall level and slope of the term structure. The predicted yield curve closely tracks the realized curve across maturities, with only modest deviations at the short end. In particular, the model correctly reproduces the smooth monotonic increase in yields from short to long maturities, indicating that the extracted level and curvature factors are well aligned with prevailing market conditions. This result is consistent with the strong out-of-sample performance observed in the RMSE analysis and suggests that the Transformer-Only architecture effectively generalizes within regimes similar to those present in the training data.

Panel (b) presents a markedly different outcome for March 2023, a period following aggressive monetary tightening in response to elevated inflation. The realized yield curve exhibits a pronounced inversion, with short-term yields substantially exceeding long-term yields. In contrast, the model predicts a downward-sloping but nearly linear curve that fails to replicate the depth and curvature of the observed inversion. While the predicted curve reflects rising short-term rates relative to longer maturities, it substantially underestimates the steepness of the inversion and overstates long-term yields.

This discrepancy highlights a key limitation of the model under sharp regime shifts. The inability to reproduce the inverted shape suggests that the extracted slope factor does not turn sufficiently negative, and that the curvature factor does not generate the pronounced mid-maturity trough observed in the realized curve. From a modeling perspective, this reflects the scarcity of deeply inverted yield curve episodes in the training sample, particularly under rapid policy tightening cycles. As a result, the model tends to extrapolate from historically dominant upward-sloping configurations, producing forecasts that resemble a smoothed or lagged version of the yield curve rather than fully adapting to abrupt structural changes.

Importantly, the figure also illustrates that the model’s errors are not uniform across maturities. The largest deviations occur at the short end of the curve, where yields are most sensitive to near-term policy expectations. This is consistent with earlier findings that the slope factor exhibits greater model-specific variability and is more difficult to estimate accurately during periods of heightened policy uncertainty. In contrast, long-maturity yields are predicted more smoothly, reflecting the model’s stronger ability to capture persistent level dynamics.

Overall, these case studies underscore the strengths and limitations of the DLNS framework. While the Transformer-Only model performs well in stable environments and captures the general structure of the yield curve, its performance deteriorates during periods of rapid regime transition and extreme slope dynamics. These findings motivate the inclusion of macroeconomic variables and alternative training strategies in subsequent experiments, with the goal of improving the model’s ability to anticipate and adapt to abrupt changes in monetary policy and inflation expectations.

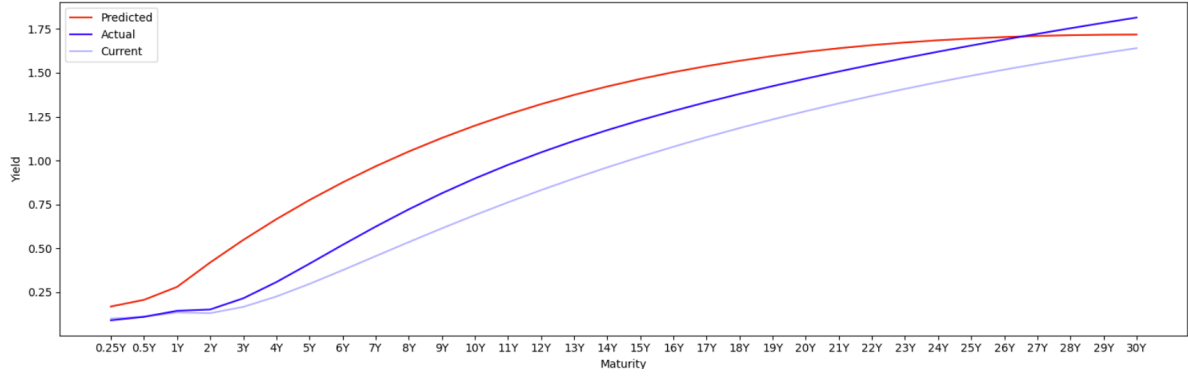
5.4 Recursive Forecasting Results and Macro Input Effect

Recursive forecasting performance is summarized in Table 2 and visualized in Figure 4. In this setup, multi-step-ahead forecasts are generated iteratively by feeding model predictions back into the input sequence. Overall, models trained without macroeconomic variables and with a shorter lookback window of six months consistently outperform those using a nine-month lookback or macro-augmented inputs.

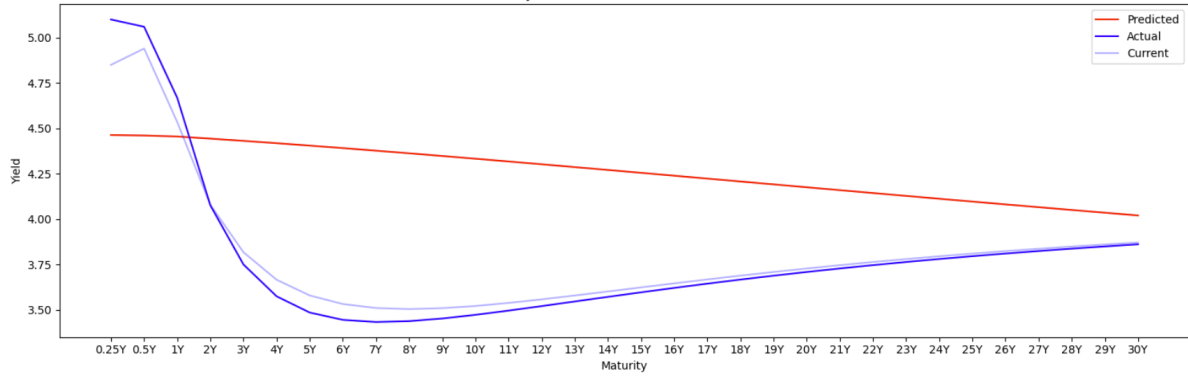
The superior performance of the shorter lookback window suggests that recent yield curve information contains the most relevant signals for near- and medium-term forecasting. Extending the lookback window appears to introduce additional noise or outdated dynamics that are less informative under changing monetary regimes. This effect is particularly pronounced for architectures that rely heavily on temporal aggregation, such as recurrent models.

The weaker performance of macro-augmented models can be attributed to a structural limitation of the current DLNS framework. While macroeconomic variables are incorporated as inputs, they are not explicitly modeled or forecasted within the system. As a result, in recursive forecasting exercises, macroeconomic inputs are held fixed at their most recent observed values and reused across forecast horizons. This treatment implicitly assumes that macroeconomic conditions remain constant over the forecast period, an assumption that is unlikely to hold during periods of rapid economic change.

Empirically, the degradation in performance of macro-augmented models becomes more pro-



(a) Predicted vs. Actual Yield Curve for October 2020



(b) Predicted vs. Actual Yield Curve for March 2023

Figure 6: Yield Curve Forecasting Results Using Transformer-Only Model (Lookback Window = 6, Without Macro Variables) *Note*: "Current" refers to the yield curve at the forecast issuance date.

nounced beyond the two-month horizon. At longer horizons, the repeated use of stale macroeconomic information may distort latent factor extraction, leading to biased slope and curvature dynamics. These findings suggest that naive inclusion of macro variables without a coherent forecasting mechanism may reduce, rather than enhance, predictive accuracy in recursive settings.

5.5 Model-wise Performance Differences

Across all experimental configurations, attention-based architectures, namely the Transformer-Only and CNN-Transformer models, consistently outperform sequence-based alternatives such as RNN and LSTM models. This pattern underscores the importance of flexible dependency modeling for yield curve forecasting. By relying on self-attention, Transformer-based models can dynamically weight information across time and maturities, allowing them to capture both short-term adjustments and long-range dependencies more effectively than recurrent structures.

In contrast, CNN-RNN and CNN-LSTM models exhibit higher error variance across maturities, particularly at shorter tenors. This suggests that recurrent architectures are more sensitive to local fluctuations at the short end of the yield curve, where yields are strongly influenced by policy expectations and exhibit greater volatility. Errors at these maturities tend to propagate through the recursive forecasting process, leading to larger deviations at longer horizons.

The CNN-Transformer model generally performs better than recurrent hybrids but remains slightly inferior to the Transformer-Only architecture. This indicates that explicitly separating cross-sectional and temporal processing may impose unnecessary constraints when self-attention mechanisms are sufficiently expressive to model both dimensions jointly. Taken together, these results highlight the advantage of architectures with a global receptive field for capturing the complex joint dynamics of the yield curve.

5.6 Model Robustness Considerations

The results presented above are based on a single training run for each model architecture. As with most deep learning methods, the estimation process involves stochastic components such as random weight initialization, mini-batch sampling, and regularization techniques. Consequently, point estimates of forecasting performance may be sensitive to the particular random seed used during training.

A more comprehensive assessment of model robustness would involve repeating each experiment across multiple random seeds and reporting average performance metrics along with measures of dispersion. Such an approach would allow for a clearer distinction between systematic architectural advantages and variation induced by stochastic optimization. Future extensions of this work will explore ensemble-based approaches and Monte Carlo dropout methods to quantify predictive uncertainty and improve the stability of out-of-sample forecasts.

Despite these limitations, the consistency of the relative ranking across architectures suggests that the main qualitative findings are robust. In particular, the superior performance of attention-based models and the limited benefits of naive macroeconomic input integration appear to be systematic features of the DLNS framework under the current experimental design.

6 Conclusion

This study examined the use of deep learning methods for modeling and forecasting the U.S. Treasury yield curve within the economic structure of the Dynamic Nelson-Siegel (DNS) framework. By replacing the linear state-space estimation commonly used in traditional DNS implementations

with nonlinear neural network encoders, the proposed approach combines economic interpretability with the flexibility of modern machine learning. This hybrid framework allows latent yield curve factors to be learned directly from data while preserving the well-established Nelson-Siegel representation of the term structure.

The primary contribution of this project lies in its systematic evaluation of alternative deep learning architectures for extracting latent yield curve factors, including level, slope, curvature, and a time-varying decay parameter. Among the models considered, the Transformer-Only architecture consistently delivered the strongest forecasting performance, particularly in configurations that relied solely on yield curve information. This result highlights the effectiveness of self-attention mechanisms in jointly capturing cross-maturity relationships and temporal dependencies without requiring explicit separation of these dimensions.

At the same time, the analysis reveals important limitations. The use of monthly data constrains the effective sample size, which is especially restrictive for high-capacity models such as Transformers. Future work could mitigate this limitation by extending the historical sample or by moving to higher-frequency data, such as daily yields, to increase the number of training observations. In addition, the results presented here are based on single training runs for each architecture. Given the stochastic nature of neural network optimization, further robustness analysis using multiple random seeds or ensemble-based approaches would be necessary to obtain more stable performance estimates.

Beyond forecasting accuracy, the proposed framework offers potential value for structural interpretation. When trained on long historical samples, the extracted latent factors provide a compact representation of yield curve dynamics that can be used to study macroeconomic regime changes. Shifts in monetary policy, inflation expectations, or financial market stress are reflected in systematic movements of the level, slope, and curvature factors, as well as changes in the decay parameter. This decomposition may prove useful for policy analysis and risk management, particularly in understanding how structural shocks propagate across maturities.

Overall, this study demonstrates that deep learning can be meaningfully integrated into economically structured yield curve models. By maintaining interpretability while enhancing flexibility and predictive performance, the DLNS framework represents a promising direction for future research at the intersection of financial economics and machine learning.

References

- [1] Francis X. Diebold and Canlin Li. Forecasting the term structure of government bond yields. *Journal of Econometrics*, 130(2):337–364, 2006.
- [2] Jens H. E. Christensen, Francis X. Diebold, and Glenn D. Rudebusch. The affine arbitrage-free class of nelson-siegel term structure models. Working Paper 2007-20, Federal Reserve Bank of San Francisco, 2010. This version: March 2010. Also published later in **Journal of Econometrics** (2011).
- [3] Sang-Heon Lee. Yield curve forecasting using deep learning nelson-siegel model. SSRN Working Paper, May 2023. Available at SSRN: <https://ssrn.com/abstract=4447541>.
- [4] Francis X. Diebold, Glenn D. Rudebusch, and S. Boragan Aruoba. The macroeconomy and the yield curve: A dynamic latent factor approach. *Journal of Econometrics*, 131(1-2):309–338, 2006.
- [5] Refet S. Gürkaynak, Brian Sack, and Jonathan H. Wright. The u.s. treasury yield curve: 1961 to the present. Finance and Economics Discussion Series 2006-28, Board of Governors of the Federal Reserve System, June 2006. Data available at: <https://www.federalreserve.gov/econresdata/research-data.htm>.
- [6] Ronald Richman and Salvatore Scognamiglio. Multiple yield curve modeling and forecasting using deep learning. *Astin Bulletin*, 54:463–494, 2024. Received: 29 February 2024; Revised: 30 June 2024; Accepted: 10 August 2024; First published online: 4 October 2024.
- [7] Refet S. Gürkaynak, Brian W. Sack, and Jonathan H. Wright. The u.s. treasury yield curve: 1961 to the present. *Journal of Monetary Economics*, 54:2291–2304, 2007.